31338 Network Servers
32520 Systems Administration

Week 5
Users, Groups and Directories

Dr Litao Yu

University of Technology, Sydney

Manage user accounts

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- Users have a username and user ID (**uid**)
  - uid is a 32-bit number
  - CentOS: UID starting from 1000: **/etc/login.defs**
- The username is a unique identifier visible by users but uid is essentially used by the OS
- Each user must belong to at least one group
- Groups have a group name and group ID (**gid**)

```
[root@192-168-70-128 ~]# cat /etc/login.defs
#
# Please note that the parameters in this configuration file control the
# behavior of the tools from the shadow-utils component. None of these
# tools uses the PAM mechanism, and the utilities that use PAM (such as the
# passwd command) should therefore be configured elsewhere. Refer to
# /etc/pam.d/system-auth for more information.
#

# *REQUIRED*
#   Directory where mailboxes reside, _or_ name of file, relative to the
#   home directory.  If you _do_ define both, MAIL_DIR takes precedence.
#   QMAIL_DIR is for Qmail
#
#QMAIL_DIR      Maildir
MAIL_DIR        /var/spool/mail
#MAIL_FILE      .mail
```

```
PASS_MAX_DAYS   99999
PASS_MIN_DAYS   0
PASS_MIN_LEN    5
PASS_WARN_AGE   7

#
# Min/max values for automatic uid selection in useradd
#
UID_MIN                  1000
UID_MAX                 60000
# System accounts
SYS_UID_MIN               201
SYS_UID_MAX               999

#
# Min/max values for automatic gid selection in groupadd
#
GID_MIN                  1000
GID_MAX                 60000
# System accounts
SYS_GID_MIN               201
SYS_GID_MAX               999
```

User management policies

- Organisations have policies for user accounts, e.g.
  - How uid's are allocated
  - What groups a user may belong to
  - How usernames will be generated
  - Security features (e.g. password aging) to be enforced
  - Whether accounting or resource limiting will be enforced

- Organisations typically require new users to complete a signup sheet:
  - indicates the commitment required of users
  - often given with acceptable use policy

/etc/passwd

- User data file of Unix-like system is **/etc/passwd**
- Readable by everybody, writable only by root

  [root@localhost named]# ls -l /etc/passwd

  -rw-r--r-- 1 root root 3210 Jul 1 11:37 /etc/passwd

  Only place where username and uid are linked
- Use **vipw** command if editing manually (does locking)
- Sample passwd file (without shadow passwords) ????

```
sysadmin@localhost:~$ tail -5 /etc/passwd
syslog:x:101:103::/home/syslog:/bin/false
bind:x:102:105::/var/cache/bind:/bin/false
sshd:x:103:65534::/var/run/sshd:/usr/sbin/nologin
operator:x:1000:37::/root:/bin/sh
sysadmin:x:1001:1001:System Administrator,,,,:/home/sysadmin:/bin/bash
```

- *username : password : uid : gid : comments : home_dir : shell*

/etc/shadow

- `/etc/passwd`
  - passwd contains the users' public information (UID, full name, home directory)
- `/etc/shadow` contains the hashed password and the password expiry data
  - Shadow entries must match users in `/etc/passwd`

  Sample `/etc/shadow` entries:

```
peter:$1$qYyXGFnx$mcagydJluWmKTCtL3f07w/:13361:0:99999:7:::
someuser:!$1$.qyZlc4K$LzRvAadGvuOAoruyckpMG.:13385:0:99999:7:::
```

  - Can be accessed by root only

username : password : lastchg : min : max : warn : inactive : expires

username: A direct match to the username in the /etc/passwd file.

password: encrypted, a blank entry indicates a password is not required to log in. System accounts have an asterisk * character in this field, new user is set to !, meaning the account is locked by default.

/etc/shadow

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```
peter:$1$qYyXGFnx$mcagydJluWmKTCtL3f07w/:13361:0:99999:7:::
someuser:$1$.qyZlc4K$LzRvAadGvuOAoruyckpMG.:13385:0:99999:7:::
daemon:*:19001:0:99999:7:::
```

username : password : lastchg : min : max : warn : inactive : expires

lastchg: The number of days (since Jan. 1, 1970*) since the password was last changed.

min: The number of days before password may be changed (0 indicates it may be changed at any time)

max: The number of days after which password *must* be changed (99999 indicates user can keep his or her password unchanged for many, many years)

warn: The number of days to warn user of an expiring password (7 for a full week)

inactive: The number of days after password expires that account is disabled

expires: The number of days since January 1, 1970 that an account has been disabled

***This is the date (date-axis) start point from 1970 AD, not 0000 AD!**

/etc/group

- `/etc/group` defines available groups
  - -rw-r--r-- 1 root root 1122 Aug 16 08:20 `/etc/group`
  - -rw-r--r-- 1 root root 449 Jul 12 2022 sysctl.conf
  - Users have one default group, but can be in many groups
    - Default group is in `/etc/passwd`
  - Users can switch groups with the `newgrp` command
    - newgrp dev #Attempt to log into the group **called dev**, current working environment remains unchanged
    - newgrp – dev #if the option – flag is given, user environment re-initialized as though he or she had just logged in, otherwise, the current environment Including current working directory, remains unchanged.
    - setgid* bits on directories more common
  - Sample `/etc/group` entries:

```
staff:x:400:someuser,me,jane,chun
someuser:x:500:
```

groupname : x : gid : userlist

* Setting the setgid permission on a directory causes files and subdirectories created within to inherit its group ownership, rather than the primary group of the file-creating process. Created subdirectories also inherit the setgid bit

Group management policies

- Different conventions:
  - User's UNIX group (system) reflects user's real-world group (e.g. peter)
  - User Private Group scheme (UPG such as RedHat) (umask 002 or 007)

UPG allows multiple users of a system to collaborate on files without any permission hassle

- Typical desires:
  - users within a group ID and can read/write one file created by another user account
  - users working on common projects can share files

Check your lab manual/handout to make this happen

- May be complicated by network filesystem access
- *ACLs also a (complex) solution to file sharing

*An access control list (ACL) contains rules that grant or deny access to certain digital environments

Add/modify/delete users

- step 1: `useradd [options] username`
  `usermod [options] username`
  `userdel [options] username`

- Step 2: `passwd username` to set the user's initial password

- Step 3: System will copy the /etc/skel to the home dir.

In principle, you can do what is required manually:
  - Create specific files in /etc/skel called skeleton files for new users.
  - Create entries in `/etc/passwd` and `/etc/shadow`
  - Create a home directory and chown it

- Many organisations use a custom shell/Perl script

Add/modify/delete users

`useradd [options] username`

The useradd command, with no options, creates a new account using default options. You can also specify the optional parameters:

useradd -s <Shell> -d <Home directory> -m -k <Skeleton directory> -g <Group> username

e.g., useradd -s /bin/bash -d /home/alex -m -g staff -g dev alex

Add/modify/delete users

`useradd [options] username`

The `-D` option for the `useradd` command is to **display** or modify the default setting for new user account on the system.

```
[root@192-168-70-128 ~]# useradd -D
GROUP=100
HOME=/home
INACTIVE=-1
EXPIRE=
SHELL=/bin/bash
SKEL=/etc/skel
CREATE_MAIL_SPOOL=yes
```

`useradd -D -s /bin/zsh`

This sets the default shell for all future users to /bin/zsh

Add/modify/delete users

- The usermod command offers options for modifying an existing user account, similar to the useradd command.
- `root@localhost:~# usermod -aG dev jane`

Explanation:

a: append the user's supplemental groups with those specified (already belong to) by the -G option.

G: Set supplemental groups to the user.

Without -a option, jane will belong to the new group dev and no other group.

Short Option	Description
-d *HOME_DIR*	Sets HOME_DIR as a new home directory for the user.
-e *EXPIRE_DATE*	Set account expiration date to EXPIRE_DATE.
-f *INACTIVE*	Set account to permit login for INACTIVE days after password expires.
-g *GROUP*	Set GROUP as the primary group.
-G *GROUPS*	Set supplementary groups to a list specified in GROUPS.
-a	Append the user's supplemental groups with those specified by the -G option.
-h	Show the help for the usermod command.
-l *NEW_LOGIN*	Change the user's login name.
-L	Lock the user account.
-s *SHELL*	Specify the login shell for the account.
-u *NEW_UID*	Specify the user's UID to be NEW_UID.
-U	Unlock the user account.

This makes jane a member of the dev group, in addition to any groups she already belongs to.

Add/modify/delete users

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- Lock and unlock accounts with 3 different ways
  1. Commands: **`passwd -l`** username and **`passwd -u username`**
  2. Put a **`*`** or **`!!`** in the password (shadow) field to lock

```
[root@192-168-70-128 ~]# grep sysadmin /etc/shadow
sysadmin:$6$6W6BI0APpSFg/Qh0$jNV4QGhlQxqzPkSUGAHrUD10v2afC.Cl7X1MS9sjg9ZFKEIuJfyZ
Ucz2Kr9NPMbV5Xr22hxcJPfvXYvL5l4QC.:19603:0:99999:7:::
[root@192-168-70-128 ~]# passwd -l sysadmin
Locking password for user sysadmin.
passwd: Success
[root@192-168-70-128 ~]# grep sysadmin /etc/shadow
sysadmin:!!$6$6W6BI0APpSFg/Qh0$jNV4QGhlQxqzPkSUGAHrUD10v2afC.Cl7X1MS9sjg9ZFKEIuJf
yZUcz2Kr9NPMbV5Xr22hxcJPfvXYvL5l4QC.:19603:0:99999:7:::
```

  3. Set a user's login shell to **`/sbin/nologin*`** or equivalent to lock

*How to do it? you can use the usermod command, along with the -s or --shell option

Example: ~# usermod -s /sbin/nologin peter

Add/modify/delete users

- userdel command is used to delete users.
  - When you delete a user account, you also need to decide whether to delete the user's home directory. The user's files may be important to the organization, and there may even be legal requirements to keep the data for a certain amount of time, so be careful not to make this decision lightly.
  - don't forget to delete/archive other files owned by user too
    - e.g. mail spool, cron jobs, at jobs, files in /tmp, /var/tmp, etc.
  - To delete the user, home directory, and mail spool as well, use the -r (recursive) option: `root@localhost:~# userdel -r jane`
- Technically, deleting a user is easy
  - the tricky part is all the policies about what to do with everything that belonged to the user

Manage user accounts

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- Set password aging if required
  - **chage** command makes users change passwords regularly
    - **chage [options] user** #The **chage** command changes the number of days between password changes and the date of the last password change.
  - users may act more insecurely if required to change too often

Short Option	Long Option	Description
-l	--list	List the account aging information
-d LAST_DAY	--lastday LAST_DAY	Set the date of the last password change to LAST_DAY
-E EXPIRE_DATE	--expiredate EXPIRE_DATE	Set account to expire on EXPIRE_DATE in the YYYY-MM-DD format
-h	--help	Show the help for the chage command
-I INACTIVE	--inactive INACTIVE	Set account to permit login for INACTIVE days after password expires
-m MIN_DAYS	--mindays MIN_DAYS	Set the minimum number of days before the password can be changed to MIN_DAYS
-M MAX_DAYS	--maxdays MAX_DAYS	Set the maximum number of days before a password should be changed to MAX_DAYS
-W WARN_DAYS	--warndays WARN_DAYS	Set the number of days before a password expires to start displaying a warning to WARN_DAYS

- Set resource limits if required
  - e.g., disk quotas, print quotas, etc.

Add/modify/delete groups


```
groupadd [options] groupname
groupmod [options] groupname
groupdel [options] groupname
```

- In principle, you can do what is required manually:
  - Create entry in `/etc/group`
  - Set a group password if required (`gpasswd` command)

- Adding users to groups:

  `usermod -g staff someuser` (set default group)

  `usermod -G staff,www,admin someuser` (add other groups)

Check registered & logged in users

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- The id command is used to print user and group information for a specified user.
- `sysadmin@localhost:~$ id`

```
uid=0(root) gid=0(root) groups=0(root)

uid=200(brian) gid=100(users) groups=100(users),1002(family)
```

- Use w, who command to see logged in users in Linux
  - direct login: TTY (teletype ie terminal, text input/output environment);
  - remote login: PTS (pseudo terminal slave) (telnet/SSH)

```
[root@192-168-70-128 ~]# w
 19:31:32 up  5:32,  2 users,  load average: 0.22, 0.10, 0.12
USER     TTY      FROM             LOGIN@   IDLE   JCPU   PCPU WHAT
root     seat0    login-           09:46    0.00s  0.00s  0.01s /usr/libex
root     tty2     tty2             09:46    9:45m  4:02   0.01s /usr/libex
[root@192-168-70-128 ~]# who
root     seat0        2023-09-03 09:46 (login screen)
root     tty2         2023-09-03 09:46 (tty2)
```

Notify users on system issues

- Terminal login
  - direct login: Teletypewriter (TTY) login via direct, physical/virtual console
  - remote login: Pseudo Terminal Slave (PTS) login via telnet/SSH
- Local login (TTY) messages: `/etc/issue`
  - displayed before login prompt on console (text-based login)
- Network login (PTS) messages: `/etc/issue.net`
  - displayed before login prompt for telnet connections
  - not displayed for SSH connections though
- Message Of The Day (TTY/PTS): `/etc/motd`
  - displayed after login for console, telnet and ssh logins
  - not typically displayed for GUI logins

Windows account admin

- Three roles for a Windows Server in a network (machine)
  - Standalone server
  - Domain controller
  - Member server
- Centralised = use internal account management
  - Security Account Manager (SAM):
- Decentralised = distributes information-processing workloads across multiple devices instead of relying on a single central server.

Workgroups -- Decentralized Mgt in Windows

- A workgroup is a logical group of computers with a peer-to-peer model and communication over LAN. Each machine can be either client or server
- Decentralized security & administration model
  - Authentication provided by the local Security Account Manager (SAM)
- Limitations
  - Users need unique accounts on each workstation
  - Users manage their own accounts (security issues)
  - Not very scalable (10-20 devices)

Domains – Centralised Mgt in Win

- A domain is a logical group of computers
  - Centralized authentication and administration
  - Authentication provided through centralized Active Directory (AD) Domain server.
  - The main purpose of a domain managed by a domain controller for an IT admin or service is to control the network, including any security issues and permissions all from one centralized location.
    - When a network administrator makes a change to one device, it will be automatically made for all the other devices that exist within the same domain.
  - Active Directory Database* can distribute amongst more than one domain controllers
  - At least one system must be configured as a domain controller

*As the primary database file, it stores objects, attributes, properties, configuration, and schema information for a local domain in the Active Directory service.

Domain -- Member Servers in Windows

- Contain an account in a domain
- Not configured as a domain controller
- Typically, be used for sharing file, prints, application, and hosting network services

Domain Controllers in Windows

- Set with Active Directory Domain service role
  - Serves user authentication requests
  - Serve queries about domain objects
- Often set up to be master DNS server and LDAP* (directory) server

*LDAP (Lightweight Directory Access Protocol) is an open and cross platform protocol used for directory services authentication. LDAP provides the communication language that can be used to communicate with other directory services servers.

Manage Users, Computers, and Groups in Windows

- Objects in Windows: users, computers, roles, etc
  - Windows admin: You can update LOCAL or DOMAIN accounts
- User accounts include:
  - Main types: administrator, power users, users, guests
- Each computer:
  - Used as domain controller (DC) for authentication and access auditing
  - Member server by registering with DC
- Groups
  - Define logical grouping of objects
  - Assign rights and permissions to groups
  - Define Admin. Templates via **group policies (GPO)**

Manage accounts

- Use GUI:
  - Start → Control Panel → User accounts
  - Server Manager → Tools → Computer Management → Local Users and Groups
  - You can view an object (user/group) properties
- Use Powershell:
  - net user command e.g., *net user /add chris*
  - wmic useraccount command

- You manage group policies via the **gpedit.msc***
  - You can edit computer or user configuration/properties

* MS Management Console (MMC) snap-in for the Local Group Policy Editor.